

MyDesign Portfolio

Element C Initial Template

Element C: Presentation and Justification of Solution Design Requirements

Design Requirement #1 (HIGH Priority)

The design must easily allow resetting and cycling between different “This or That” questions.

In the video where the librarian explained the “this or that” device, she mentioned that the question would cycle regularly, so some way to change out the question and reset the votes would be a top priority of our design.

Design Requirement #2 (HIGH Priority)

The design must be interactive and engage the kids in the library.

The point of this device is so that kids can pass time in line, while also being a starter for conversations. It must be fun and not lose interest with time as the novelty disappears.

Design Requirement #3 (HIGH Priority)

The design must be durable, and withstand rough treatment from middle schoolers for a few months.

Durability is key to the success, because we will not be there to repair the device if it breaks. If something gets broken and makes the device difficult to operate, our design can't be used anymore.

Design Requirement #4 (HIGH Priority)

The design must function properly and be easy to operate for middle schoolers.

This is pretty much a given. Are the pom-poms or other voting items used easily accessible? Is it easy to vote? The device should be intuitive to use. This is lower on the list because it's pretty difficult to make a device like this hard to use.

Design Requirement #5 (HIGH Priority)

The design must be safe and shouldn't harm the middle schoolers.

The librarian stated that she wanted the device to be sturdy and not fall off the desk. She also mentioned that it shouldn't be made of glass, as that could be a safety issue. Any sharp edges or things like that shouldn't be in the final design.

Design Requirement #6 (LOW Priority)

This device should be made with available materials at a reasonable cost, with easy-to-follow instructions for putting together if it needs to be assembled.

Mrs. Thompson mentioned that if the device was to be 3d printed and had multiple parts, it should have instructions for putting it together. The design also should be made with readily available materials, like cardboard, plastic, 3d printed, etc..